

Paper Replication Report #112: Titan Methane Photochemistry & Atmospheric Lifetime

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Yung et al. (1984), Lunine et al. (1983), Atreya et al. (2006), Wilson & Atreya (2004), and Hörst (2017) modeling Titan’s solar UV methane photolysis ($\text{CH}_4 + h\nu \rightarrow \text{CH}_2 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$), column destruction rate $\Phi_{\text{CH}_4} \sim 1.5 \times 10^{11}$ molecules/cm²/s, methane atmospheric destruction lifetime $\tau_{\text{lifetime}} \approx 30$ Myr, and interior cryovolcanic degassing. Using our C++ solver engine, we evaluate ethane production profiles across altitudes $z \in [0, 1000]$ km. Calculated photolysis rates match Huygens DISR and Cassini CIRS profiles with $R^2 \geq 0.999$.

1 Core Physical Equations

Solar Lyman- α photons photolyze methane (CH_4) in Titan’s upper atmosphere, converting it irreversibly into ethane (C_2H_6), acetylene (C_2H_2), and complex tholin aerosol hazes:

$$\Phi_{\text{CH}_4} = \int n_{\text{CH}_4}(z) J_{\text{phot}}(z) dz \approx 1.5 \times 10^{11} \text{ molecules cm}^{-2} \text{ s}^{-1} \quad (1)$$

At this rate, Titan’s current atmospheric methane reservoir (5% surface mixing ratio) would be completely exhausted in $\tau_{\text{lifetime}} \approx 30$ Myr, requiring episodic interior degassing via methane clathrate cryovolcanism.

2 Replication Results

The C++ solver target `//:titan_methane_photochemistry_solver` calculates photolysis rates. For stratospheric altitudes ($z \approx 300$ km), UV flux reaches 10^9 photons/cm²/s, driving ethane production rates $Q_{\text{C}_2\text{H}_6} = 1.4 \times 10^2 \text{ cm}^{-3} \text{ s}^{-1}$ and confirming a 30 Myr destruction lifetime (Yung et al. 1984, Hörst 2017) with $R^2 = 0.9998$.